

Jonathan Nicholas, Ph.D.

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Employment History

- 2023 – now **Postdoctoral Researcher**, New York University
Advisor: Marcelo Mattar
- 2015 – 2017 **Research Software Developer**, Stanford Cognitive and Systems Neuroscience Lab

Education

- 2017 – 2023 **Ph.D., Columbia University**, Psychology (Cognitive Neuroscience)
Advisor: Daphna Shohamy
- 2011 – 2015 **B.Sc. with Honors, Brown University**, Cognitive Neuroscience

Grants and Fellowships

- 2025-2027 NSF SBE Postdoctoral Research Fellowship (\$160,000)
"Exploring the role of episodic memory in adaptive decision making"
- 2021-2022 Leo Rubinstein Endowed Fellowship
- 2017-2020 NSF Graduate Research Fellowship

Honors and Awards

- 2024 Best Talk Award, Society for Neuroeconomics
NYU Postdoctoral Travel Award
- 2022 Edward E. Smith Memorial Award in Cognitive Neuroscience
- 2015 Kling Premium in Psychology
Election to Sigma Xi
- 2014 Karen T. Romer Undergraduate Teaching and Research Award
1st Prize Brown Institute for Brain Sciences Neural Decoding Competition

Research

Journal Articles

- 1 **Nicholas, J.** and Mattar, M.G., "Episodic memory facilitates flexible decision making via access to detailed events," *Nature Human Behaviour (In Press)*, 2025. 🌐 DOI: 10.1101/2025.03.13.643066.
- 2 **Nicholas, J.**, Daw, N.D., and Shohamy, D., "Proactive and reactive construction of memory-based preferences," *Nature Communications*, 2025. 🌐 DOI: 10.1038/s41467-025-56183-4.
- 3 Montaser-Kouhsari, L. *, **Nicholas, J. ***, Gerraty, R.T., and Shohamy, D., "Differentiating reinforcement learning and episodic memory in value-based decisions in parkinson's disease," *Journal of Neuroscience*, 2025, ***Denotes co-first author.** 🌐 DOI: 10.1523/JNEUROSCI.0911-24.2025.
- 4 **Nicholas, J.**, Amlang, C.J., Lin, C.Y., Desai, N., Montaser-Kouhsari, L., Kuo, S.H., and Shohamy, D., "The role of the cerebellum in learning to predict reward: Evidence from cerebellar ataxia," *The Cerebellum*, 2024. 🌐 DOI: 10.1007/s12311-023-01633-2.

- 5 Luo, X., Rechar, A., Sun, G., N. Yanez, F., Yilmaz, B., Lee, K., Cohen, A.O, Borghesani, V., Pashkov, A., Marinazzo, D., **Nicholas, J.**, ..., and Love, B.C., "Large language models surpass human experts in predicting neuroscience results," *Nature Human Behaviour*, 2024. [DOI: 10.1038/s41562-024-02046-9](#).
- 6 **Nicholas, J.**, Daw, N.D., and Shohamy, D., "Uncertainty alters the balance between incremental learning and episodic memory," *eLife*, 2022. [DOI: 10.7554/eLife.81679](#).
- 7 Grossman, I., Rotella, A., Hutcherson, C.A., ..., **Nicholas, J.**, ..., and Wilkening, T., "Insights into accuracy of social scientists' forecasts of societal change," *Nature Human Behaviour*, 2022. [DOI: 10.1038/s41562-022-01517-1](#).
- 8 Chen, L., Iuculano, T., Mistry, P., **Nicholas, J.**, Zhang, Y., and Menon, V., "Linear and nonlinear profiles of weak behavioral and neural differentiation between numerical operations in children with math learning difficulties," *Neuropsychologia*, 2021. [DOI: 10.1016/j.neuropsychologia.2021.107977](#).
- 9 Iuculano, T., Padmanabhan, A., Chen, L., **Nicholas, J.**, Mitsven, S., de los Angeles, C., and Menon, V., "Neural correlates of cognitive variability in childhood autism and relation to heterogeneity in decision-making dynamics," *Developmental Cognitive Neuroscience*, 2020. [DOI: 10.1016/j.dcn.2020.100754](#).
- 10 Dimsdale-Zucker, H.* and **Nicholas, J.***, "Is spatial context privileged in the neural representation of events?" *Journal of Neuroscience*, 2018, *Denotes co-first author. [DOI: 10.1523/JNEUROSCI.0949-18.2018](#).
- 11 Taghia, J., Cai, W., Ryali, S., Kochalka, J., **Nicholas, J.**, Chen, T., and Menon, V., "Uncovering hidden brain state dynamics that regulate performance and decision-making during cognition," *Nature Communications*, 2018. [DOI: 10.1038/s41467-018-04723-6](#).
- 12 Ryali, S., Supekar, K., Chen, T., Kochalka, J., Cai, W., **Nicholas, J.**, Padmanabhan, A., and Menon, V., "Temporal dynamics and developmental maturation of salience, default and central-executive network interactions revealed by variational bayes hidden markov modeling," *PLOS Computational Biology*, 2016. [DOI: 10.1371/journal.pcbi.1005138](#).

Conference Proceedings

- 1 **Nicholas, J.** and Mattar, M.G., "Humans use episodic memory to access features of past experience for flexible decision making," in *46th Proceedings of the Annual Meeting of the Cognitive Science Society*, Rotterdam, The Netherlands, 2024. [URL: https://escholarship.org/uc/item/9x22d800](https://escholarship.org/uc/item/9x22d800).

Presentations

Invited and Selected Talks

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| 2025 | UCL Max Planck Computational Psychiatry Seminar Series , Virtual |
| | Johns Hopkins Ocular Motor & Vestibular Lecture Series , Virtual |
| 2024 | Annual Meeting of the Cognitive Science Society , Rotterdam, Netherlands |
| | Society for Neuroscience , Chicago, IL |
| | Society for Neuroeconomics , Cascais, Portugal |
| 2022 | 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making , Providence, RI |
| | Society for Neuroeconomics , Arlington, VA |
| 2019 | Columbia Interdisciplinary Decision Making Meeting , New York, NY |
| | Manhattan Area Memory Meeting , Princeton, NJ |

Presentations (continued)

Posters

- 2025 **6th Multidisciplinary Conference on Reinforcement Learning and Decision Making**, Dublin, Ireland
- 2024 **Cognitive Computational Neuroscience**, Boston, MA
- 2022 **Society for Neuroscience**, San Diego, CA
- Neurobiology of Reward and Decision Making**, Lake Arrowhead, CA
- 18th Annual Context and Episodic Memory Symposium**, Philadelphia, PA
- International Congress of Parkinson's Disease and Movement Disorders**, Madrid, Spain
- International Congress for Ataxia Research**, Dallas, TX
- 2019 **Society for Neuroscience**, Chicago, IL
- Cognitive Neuroscience Society**, San Francisco, CA
- 2016 **Fourth Annual Flux Congress**, St. Louis, MO
- 2014 **Brown Summer Research Symposium**, Providence, RI

Academic Service

Organizing

- 2022 **Application Support Mentor**, NYU Application Support Group
- 2025-2026 **Organizer**, NYU ConCats Seminar Series
- 2024-2025 **Organizer**, NYU ConCats Seminar Series
- 2022 **Research Mentor**, Columbia Summer Internship Program in Psychological Science
- 2020-2021 **Instructor**, Columbia University Introduction to Programming Bootcamp
- 2020 **Organizer**, Columbia Interdisciplinary Decision Making Meeting
- Scientific Computing Support Staff**, Columbia Psychology Department
- 2019 **Organizer**, Columbia University Introduction to Programming Bootcamp
- Organizer**, Manhattan Area Memory Meeting

Teaching

- 2022 **Teaching Fellow**, Science of Psychology, Columbia University
- Teaching Fellow**, Cognitive Neuroscience, Columbia University
- 2020 **Teaching Fellow**, Statistics for Behavioral Scientists, Columbia University
- 2019 **Teaching Fellow**, Cognitive Neuroscience, Columbia University
- 2018 **Teaching Fellow**, Experimental Methods, Columbia University
- 2017 **Teaching Fellow**, Behavioral Neuroscience, Columbia University
- 2015 **Teaching Assistant**, Computational Cognitive Science, Brown University

Mentoring

Undergraduate Honors Theses

- 2019-2020 Nicole van Amerongen
- 2018-2020 Jessica Hecht

Academic Service (continued)

Undergraduate Research Assistants

2025-2026	Dariusz Nashat
2023-2025	Yifei Deng
2022	Annie Xu
	Sukriti Gupta
2020	Jesse Eiseman
2019	Natasha King

High School Research Assistants

2022	Andy Feng
	Hitomi Nakamura
2021-2022	Pradnya Rajalakshmi
2021	Loc Nguyen
	Brad Ji

Peer Review

Nature Communications, eLife, PLOS Computational Biology, Journal of Experimental Psychology: General, Cognitive Science, Scientific Reports, PLOS One

Technical Skills

Research Methods

Task design, Bayesian modeling, Reinforcement learning, Eyetracking, fMRI, MEG

Programming Languages

Proficient: Python, Matlab, Javascript

Competent: Stan, R, Unity/C, Bash, HTML/CSS